STAT 443 Project

```r
library(forecast)
```

```
Warning: package 'forecast' was built under R version 4.4.3
```

```r
# Function for variogram as defined in
# Bisgaard and Kulahci (2011). Time Series Analysis and Forecasting by Example, Wiley
variogram = function(y, lagmax=10, iprint=F)
{ G = rep(1,lagmax)
n = length(y)
if(lagmax>n) { lagmax = n-2 }
y1 = y[-1]; y2 = y[-n]
d1 = y1-y2; denom = var(d1)
for(k in 2:lagmax)
{ y1 = y[(k+1):n]; y2 = y[1:(n-k)]
dk = y1-y2
numer = var(dk)
G[k] = numer/denom
}
# H is the variogram assuming a stationary time series
H = rep(1,lagmax)
ac = c(acf(y,plot=F,lag.max=lagmax)$acf)
H = (1-ac[-1])/(1-ac[2])
if(iprint)
{ print(cbind(G,H))
mx = max(cbind(G,H))
matplot(1:lagmax,cbind(G,H),ylim=c(0,mx),ylab="variogram", xlab="lag")
abline(h=1)
}
list(G=G, H=H)
}
```

```r
persist_fc <- function(train, holdout, iprint = FALSE) {
  ntrain <- length(train)
  nh <- length(holdout)
  sse <- 0
  fc_vec <- numeric(nh)
  fc <- train[ntrain]
  yt <- holdout[1]
  fc_vec[1] <- fc
  fcerror <- yt - fc
  sse <- sse + (fcerror^2)
  for (i in 2:nh) {
    yt <- holdout[i]
    fc <- holdout[i-1]
    fc_vec[i] <- fc
    fcerror <- yt - fc
    sse <- sse + (fcerror^2)
  }
  return(list(rmse = sqrt(sse/nh), fc = fc_vec))
}

# Linear in the previous observation
# b0,b1 from least squares or another estimation method
linreg_fc = function(train_last, holdout, b0,b1)
{ nh = length(holdout)
  previous = c(train_last, holdout[-nh])
  fc_vec = b0+b1*previous
  rmse = sqrt(mean((holdout-fc_vec)^2))
  return(list(rmse=rmse, fc=fc_vec))
}

# Simple Exponential Smoothing
esm_fc <- function(train, holdout, alpha, level, iprint = FALSE) {
  nh <- length(holdout)
  sse <- 0
  fc_vec <- numeric(nh)
  L <- level
  for (i in 1:nh) {
    fc <- L
    fc_vec[i] <- fc
    yt <- holdout[i]
    fcerror <- yt - fc
    sse <- sse + (fcerror^2)
```

```r
    L <- alpha * yt + (1 - alpha) * L
  }
  return(list(rmse = sqrt(sse / n_holdout), fc = fc_vec))
}
```

```r
plot_forecast <- function(holdout, fc, model_name = "Model") {

  x <- seq_along(holdout)

  plot(x, holdout, type = "l", col = "black", lwd = 2,
       main = paste("Forecast vs Actual -", model_name),
       ylab = "Value", xlab = "Time")

  lines(x, fc, col = "red", lwd = 2)

  legend("topleft",
         legend = c("Actual", "Forecast"),
         col = c("black", "red"),
         lty = 1, lwd = 2)
}
```

```r
arima_rolling_fc <- function(fitted_model, holdout, iprint = FALSE) {
  nh <- length(holdout)
  sse <- 0
  fc_vec <- numeric(nh)

  # Extract the training data from the fitted model
  train_data <- fitted_model$x
  current_data <- train_data

  for (i in 1:nh) {
    # Forecast one step ahead using current model
    fc <- forecast(Arima(current_data, model = fitted_model), h = 1)$mean[1]
    fc_vec[i] <- fc

    yt <- holdout[i]
    fcerror <- yt - fc
    sse <- sse + (fcerror^2)

    # Expand training data with actual observed value
    current_data <- c(current_data, yt)
  }
```

```
  return(list(rmse = sqrt(sse / nh), fc = fc_vec))
}
```

Load Data

```
# Load Spot Crude Oil Price
#crude_oil <- read.csv("data/WTISPLC.csv")
#crude_oil_ts <- ts(crude_oil$WTISPLC,start=c(1975,1,1),freq=12)

#head(crude_oil_ts)

# Load Spot Crude Oil Price
crude_oil <- read.csv("data/bloomberg_oil_spot_monthly.tsv", header = TRUE, sep = "\t")
crude_oil$Date <- as.Date(crude_oil$Date)
crude_oil <- crude_oil[order(crude_oil$Date), ]

crude_oil_ts <- ts(
  crude_oil$Last.Price,
  start = c(as.numeric(format(min(crude_oil$Date), "%Y")),
            as.numeric(format(min(crude_oil$Date), "%m"))),
  frequency = 12
)

head(crude_oil_ts)
```

```
     Jan Feb Mar Apr May Jun
1970 1.8 1.8 1.8 1.8 1.8 1.8
```

Split Training - Holdout

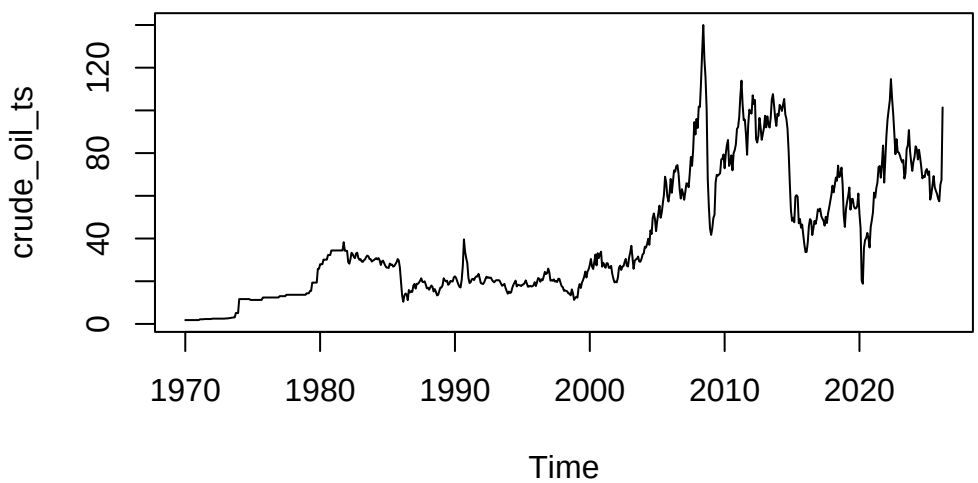
```
crude_oil_train_ts = window(crude_oil_ts, end = c(2015,12))
crude_oil_holdout_ts = window(crude_oil_ts, start = c(2016,1))

d_crude_oil_train_ts = diff(crude_oil_train_ts)
d_crude_oil_holdout_ts = diff(crude_oil_holdout_ts)

dl_crude_oil_train_ts = diff(log(crude_oil_train_ts))
dl_crude_oil_holdout_ts = diff(log(crude_oil_holdout_ts))
```

```r
end(d_crude_oil_holdout_ts)
```

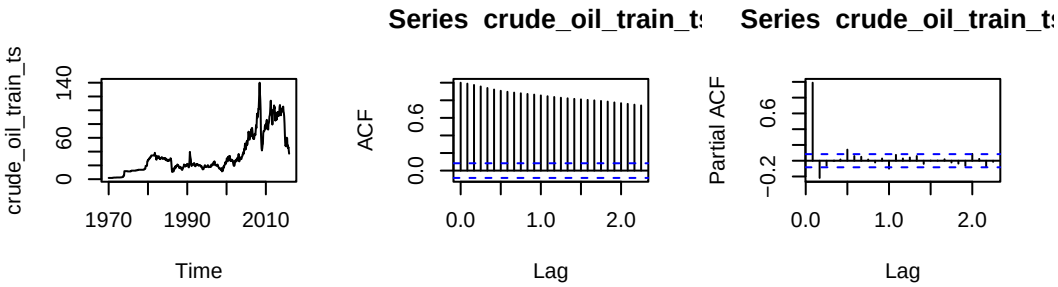
```
[1] 2026    3
```

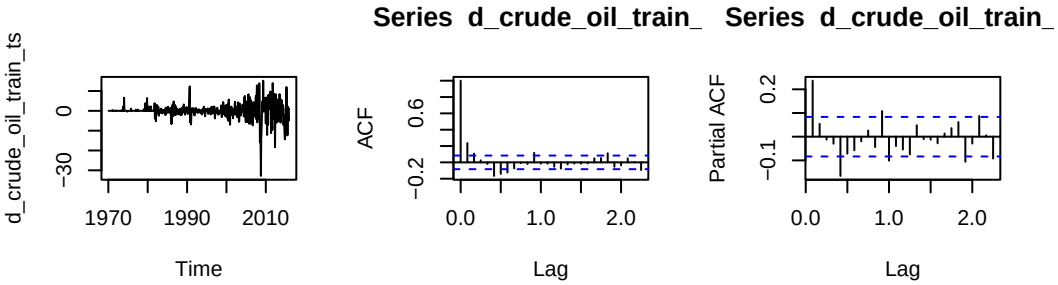
##Diagnostic Plots

```r
plot(crude_oil_ts)
```

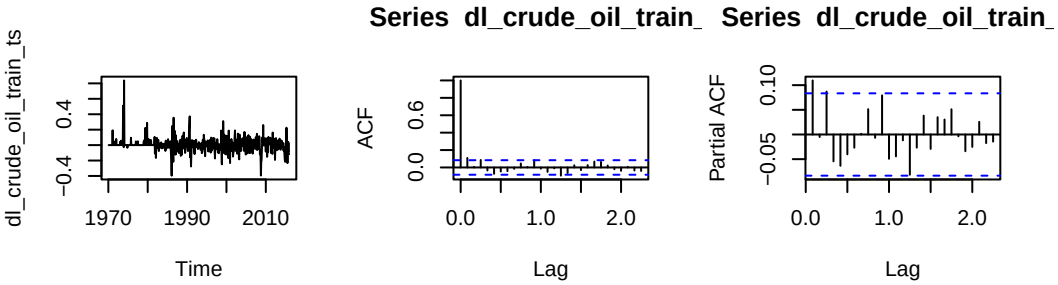



```r
par(mfrow = c(2,3))

plot.ts(crude_oil_train_ts)
acf(crude_oil_train_ts)
pacf(crude_oil_train_ts)

plot.ts(d_crude_oil_train_ts)
acf(d_crude_oil_train_ts)
pacf(d_crude_oil_train_ts)
```



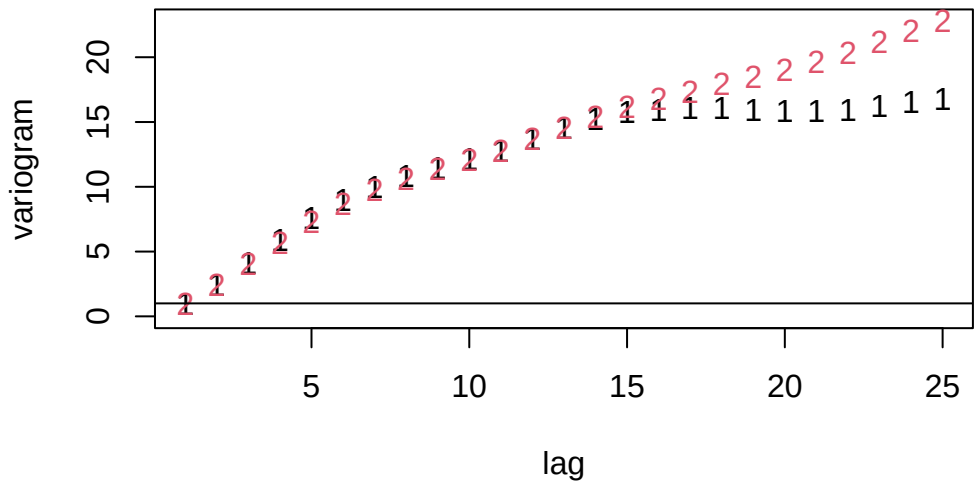



```
plot.ts(dl_crude_oil_train_ts)
acf(dl_crude_oil_train_ts)
pacf(dl_crude_oil_train_ts)
```



```r
variogram(crude_oil_train_ts, lagmax=25, iprint=T)
```

```
              G         H
 [1,]  1.000000  1.000000
 [2,]  2.474550  2.423892
 [3,]  4.160774  4.041913
 [4,]  5.907120  5.704641
 [5,]  7.611013  7.336583
 [6,]  8.991103  8.672080
 [7,] 10.040564  9.764355
 [8,] 10.837542 10.637409
 [9,] 11.489509 11.378351
[10,] 12.175925 12.096999
[11,] 12.817142 12.781864
[12,] 13.702954 13.673721
[13,] 14.542229 14.541946
[14,] 15.277551 15.388386
[15,] 15.773264 16.149354
[16,] 15.980871 16.780922
[17,] 16.067764 17.372521
[18,] 16.088081 17.939390
[19,] 15.955455 18.483965
[20,] 15.837382 19.005429
[21,] 15.866542 19.614193
[22,] 15.973822 20.320443
[23,] 16.293525 21.226932
[24,] 16.581299 22.035407
[25,] 16.774669 22.778202
```

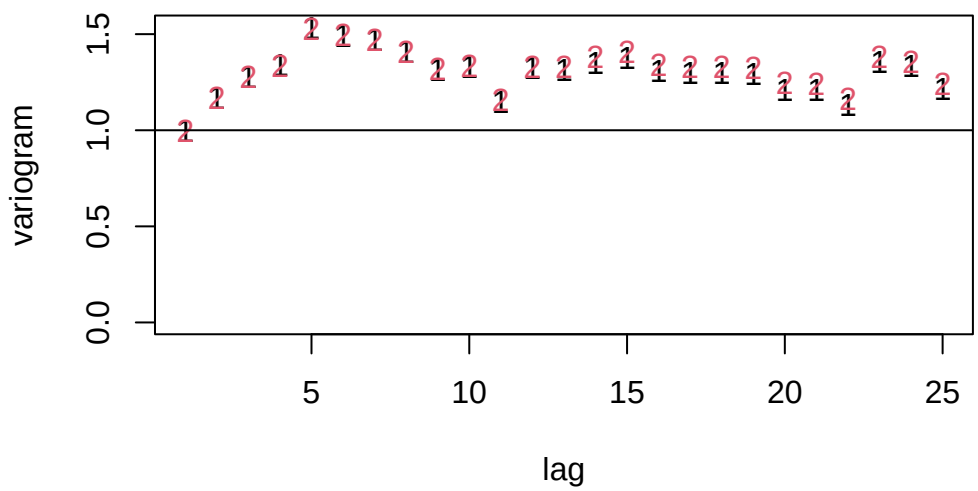



```
$G
 [1]  1.000000  2.474550  4.160774  5.907120  7.611013  8.991103 10.040564
 [8] 10.837542 11.489509 12.175925 12.817142 13.702954 14.542229 15.277551
[15] 15.773264 15.980871 16.067764 16.088081 15.955455 15.837382 15.866542
[22] 15.973822 16.293525 16.581299 16.774669

$H
 [1]  1.000000  2.423892  4.041913  5.704641  7.336583  8.672080  9.764355
 [8] 10.637409 11.378351 12.096999 12.781864 13.673721 14.541946 15.388386
[15] 16.149354 16.780922 17.372521 17.939390 18.483965 19.005429 19.614193
[22] 20.320443 21.226932 22.035407 22.778202
```

```r
variogram(d_crude_oil_train_ts, lagmax=25, iprint=T)
```

```
             G        H
 [1,] 1.000000 1.000000
 [2,] 1.169768 1.169299
 [3,] 1.279303 1.276278
 [4,] 1.338888 1.334470
 [5,] 1.535414 1.527107
 [6,] 1.493402 1.494720
 [7,] 1.472819 1.471748
```

```
 [8,] 1.409755 1.406940
 [9,] 1.313565 1.320756
[10,] 1.330906 1.335785
[11,] 1.150671 1.156871
[12,] 1.327376 1.329675
[13,] 1.317000 1.330169
[14,] 1.353634 1.379775
[15,] 1.378493 1.410469
[16,] 1.308612 1.342014
[17,] 1.297261 1.329074
[18,] 1.296515 1.330207
[19,] 1.294349 1.326229
[20,] 1.211680 1.244800
[21,] 1.208176 1.239542
[22,] 1.131201 1.163549
[23,] 1.357455 1.380179
[24,] 1.336983 1.358228
[25,] 1.217892 1.244461
```

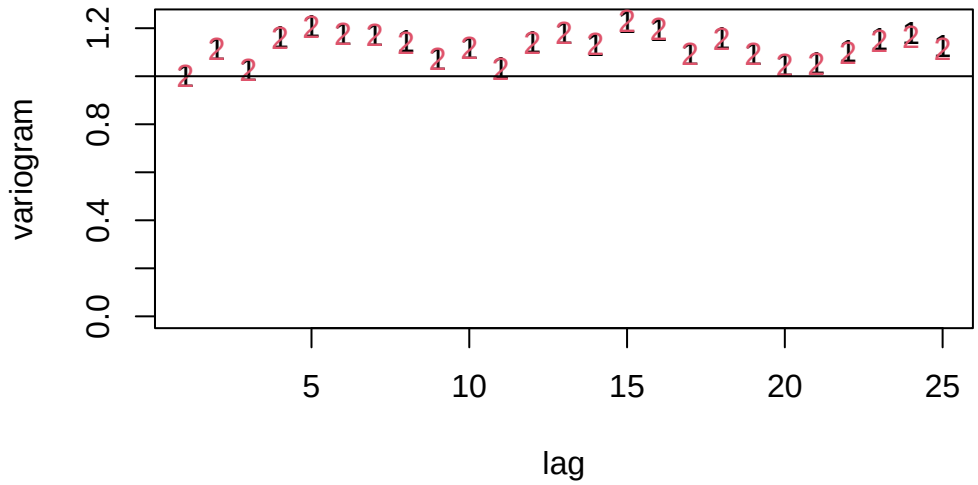



```
$G
 [1] 1.000000 1.169768 1.279303 1.338888 1.535414 1.493402 1.472819 1.409755
 [9] 1.313565 1.330906 1.150671 1.327376 1.317000 1.353634 1.378493 1.308612
```

```
[17] 1.297261 1.296515 1.294349 1.211680 1.208176 1.131201 1.357455 1.336983
[25] 1.217892

$H
 [1] 1.000000 1.169299 1.276278 1.334470 1.527107 1.494720 1.471748 1.406940
 [9] 1.320756 1.335785 1.156871 1.329675 1.330169 1.379775 1.410469 1.342014
[17] 1.329074 1.330207 1.326229 1.244800 1.239542 1.163549 1.380179 1.358228
[25] 1.244461
```

```r
variogram(dl_crude_oil_train_ts, lagmax=25, iprint=T)
```

```
             G        H
 [1,] 1.000000 1.000000
 [2,] 1.115842 1.115223
 [3,] 1.028348 1.026259
 [4,] 1.165673 1.161732
 [5,] 1.211007 1.204701
 [6,] 1.176588 1.175070
 [7,] 1.174331 1.170752
 [8,] 1.146644 1.141335
 [9,] 1.071581 1.070979
[10,] 1.120124 1.116991
[11,] 1.033230 1.029867
[12,] 1.145333 1.138939
[13,] 1.179121 1.179671
[14,] 1.130358 1.134822
[15,] 1.227213 1.228985
[16,] 1.194830 1.195713
[17,] 1.092858 1.095065
[18,] 1.158625 1.157413
[19,] 1.093897 1.092847
[20,] 1.048128 1.046805
[21,] 1.055740 1.052300
[22,] 1.104920 1.097634
[23,] 1.157627 1.146347
[24,] 1.179861 1.166222
[25,] 1.127467 1.114371
```




```
$G
 [1] 1.000000 1.115842 1.028348 1.165673 1.211007 1.176588 1.174331 1.146644
 [9] 1.071581 1.120124 1.033230 1.145333 1.179121 1.130358 1.227213 1.194830
[17] 1.092858 1.158625 1.093897 1.048128 1.055740 1.104920 1.157627 1.179861
[25] 1.127467

$H
 [1] 1.000000 1.115223 1.026259 1.161732 1.204701 1.175070 1.170752 1.141335
 [9] 1.070979 1.116991 1.029867 1.138939 1.179671 1.134822 1.228985 1.195713
[17] 1.095065 1.157413 1.092847 1.046805 1.052300 1.097634 1.146347 1.166222
[25] 1.114371
```

We first try ARIMA

ARMA on difference

```
dwti_arma = auto.arima(d_crude_oil_train_ts, seasonal=F, stationary=T)
print(dwti_arma)
```

```
Series: d_crude_oil_train_ts
```

```
ARIMA(1,0,0) with zero mean

Coefficients:
         ar1
      0.2367
s.e.  0.0414

sigma^2 = 14.55:  log likelihood = -1518.97
AIC=3041.94   AICc=3041.96   BIC=3050.56
```

Rolling 1-step forward

```r
dwti_arma_rolling = arima_rolling_fc(dwti_arma, d_crude_oil_holdout_ts)
dwti_arma_rolling$rmse
```

```
[1] 6.857281
```

ARMA on log return

```r
dlwti_arma = auto.arima(dl_crude_oil_train_ts, seasonal=F, stationary=T)
print(dlwti_arma)
```

```
Series: dl_crude_oil_train_ts
ARIMA(0,0,1) with zero mean

Coefficients:
         ma1
      0.1155
s.e.  0.0433

sigma^2 = 0.008728:  log likelihood = 524.88
AIC=-1045.75   AICc=-1045.73   BIC=-1037.13
```

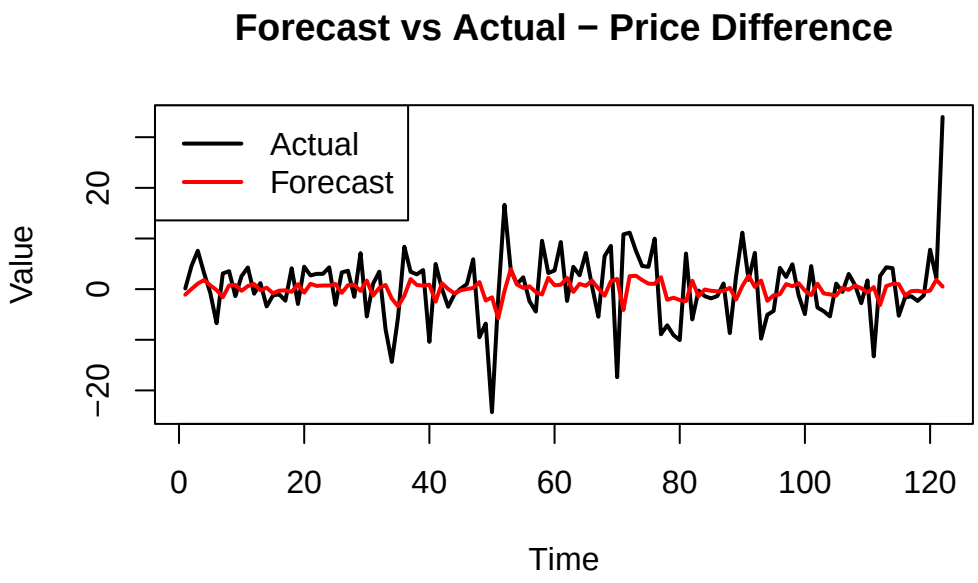
Rolling 1-step forward

```r
dlwti_arma_rolling = arima_rolling_fc(dlwti_arma, dl_crude_oil_holdout_ts)
dlwti_arma_rolling$rmse
```

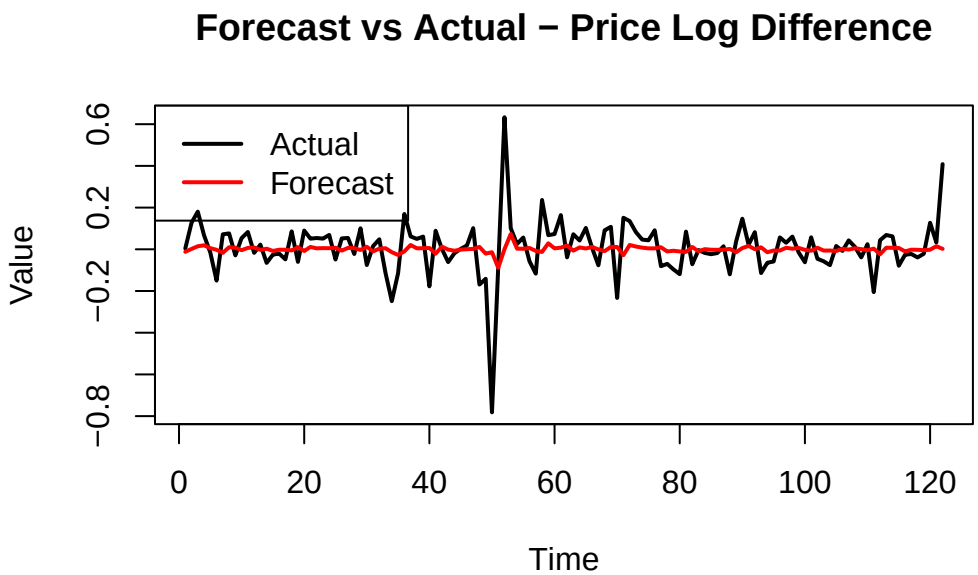
```
[1] 0.1295432
```

Plots

```r
plot_forecast(d_crude_oil_holdout_ts, dwti_arma_rolling$fc, "Price Difference")
```




```r
plot_forecast(dl_crude_oil_holdout_ts, dlwti_arma_rolling$fc, "Price Log Difference")
```



We then introduce extrogenous variables and split Train-Holdout

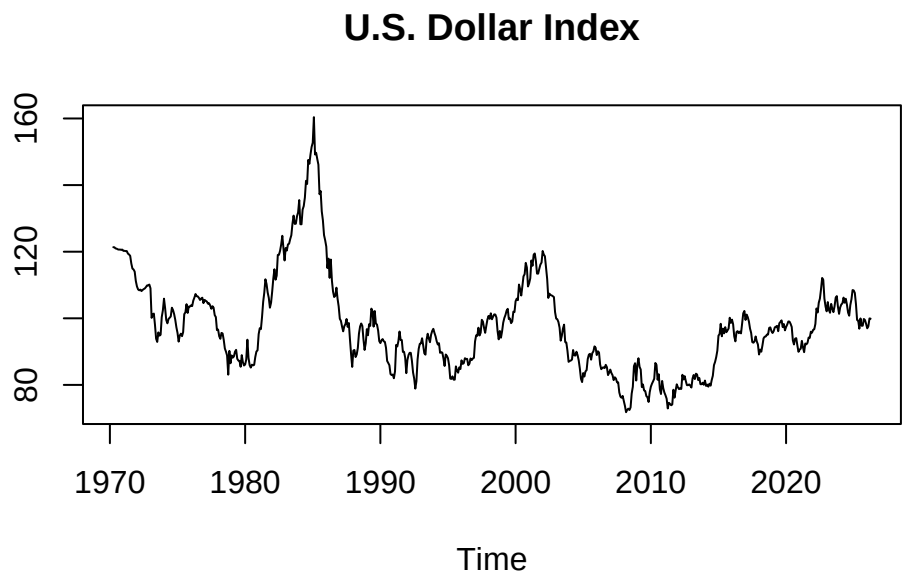
```
#usd_idx = read.csv("data/TWEXBGSMTH.csv")
#usd_idx = read.csv("data/dxy_monthly.csv")
#ind_prod = read.csv("data/INDPRO.csv")
#opec_prod = read.csv("data/opcrtotl_monthly.tsv")

#usd index
usd_idx <- read.csv("data/dxy_monthly.tsv", header = TRUE, sep = "\t")
usd_idx$Date <- as.Date(usd_idx$Date)
usd_idx <- usd_idx[order(usd_idx$Date), ]

usd_idx_ts <- ts(
  usd_idx$Last.Price,
  start = c(as.numeric(format(min(usd_idx$Date), "%Y")),
            as.numeric(format(min(usd_idx$Date), "%m"))),
  end = c(2026,04),
  frequency = 12
)
usd_idx_train_ts = window(usd_idx_ts, start = c(1975,1), end = c(2015,12))
```

```r
usd_idx_holdout_ts = window(usd_idx_ts, start = c(2016,1), end = c(2026,1))

#Industrial production
ind_prod = read.csv("data/INDPRO.csv")
ind_prod$observation_date <- as.Date(ind_prod$observation_date)
ind_prod_ts <- ts(ind_prod$INDPRO, start = c(1919,1), end = c(2026,2), frequency = 12)
ind_prod_train_ts = window(ind_prod_ts, start = c(1975,1), end = c(2015,12))
ind_prod_holdout_ts = window(ind_prod_ts, start = c(2016,1), end = c(2026,1))

# OPEC oil production
opec_prod <- read.csv("data/opcrtotl_monthly.tsv", header = TRUE, sep = "\t")
opec_prod$Date <- as.Date(opec_prod$Date)
opec_prod <- opec_prod[order(opec_prod$Date), ]

opec_prod_ts <- ts(
  opec_prod$Last.Price,
  start = c(as.numeric(format(min(opec_prod$Date), "%Y")),
            as.numeric(format(min(opec_prod$Date), "%m"))),
  end = c(2026,03),
  frequency = 12
)
opec_prod_train_ts = window(opec_prod_ts, start = c(1975,1), end = c(2015,12))
opec_prod_holdout_ts = window(opec_prod_ts, start = c(2016,1), end = c(2026,1))

# World MSCI Oil/Energy Company Index
#TO DO



head(usd_idx)
```

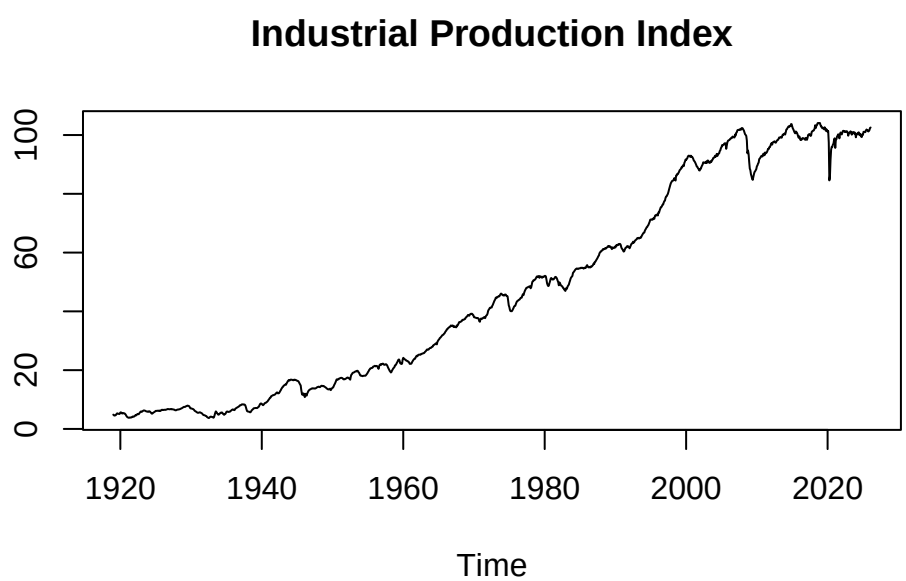
```
          Date Last.Price
673 1970-04-30    121.360
672 1970-05-29    121.322
671 1970-06-30    120.978
670 1970-07-31    120.904
669 1970-08-31    120.704
668 1970-09-30    120.620
```

```r
#usd_ts <- ts(usd_idx$Last.Price, start = c(1975,1), frequency = 12)
#ind_ts <- ts(ind_prod$INDPRO, start = c(1975,1), frequency = 12)
#opec_ts <- ts(opec_prod$Last.Price, start = c(1975,1), frequency = 12)
```

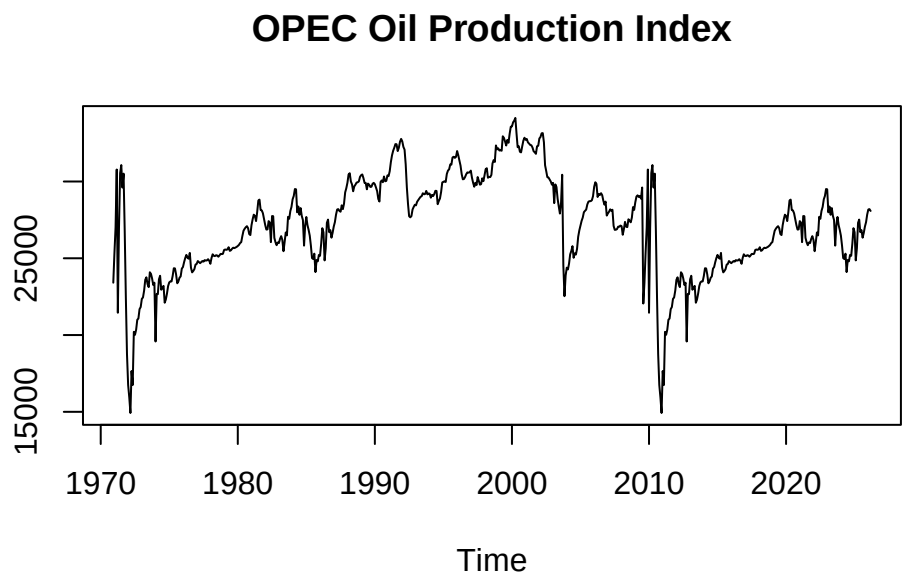
```r
plot(usd_idx_ts, main="U.S. Dollar Index", ylab="")
```




```r
plot(ind_prod_ts, main="Industrial Production Index", ylab="")
```



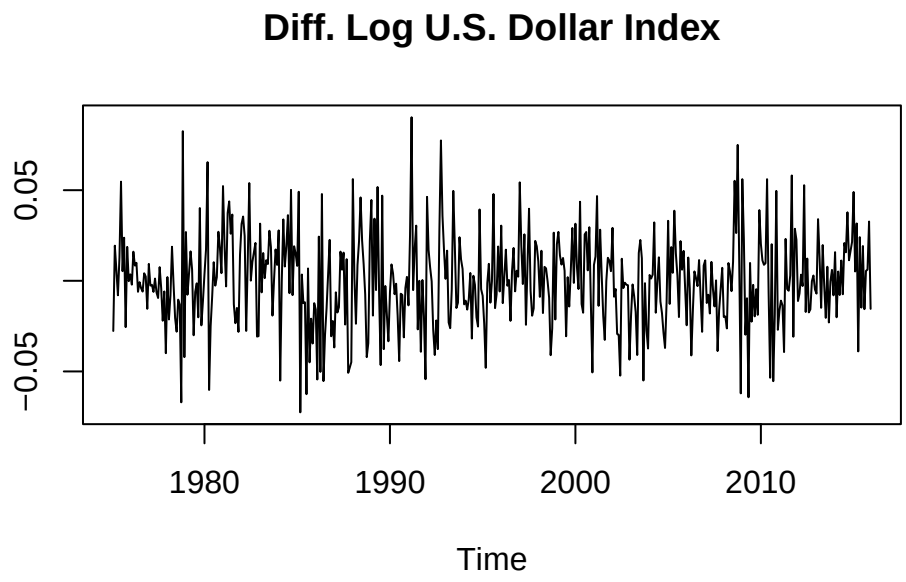
```r
plot(opec_prod_ts, main="OPEC Oil Production Index", ylab="")
```



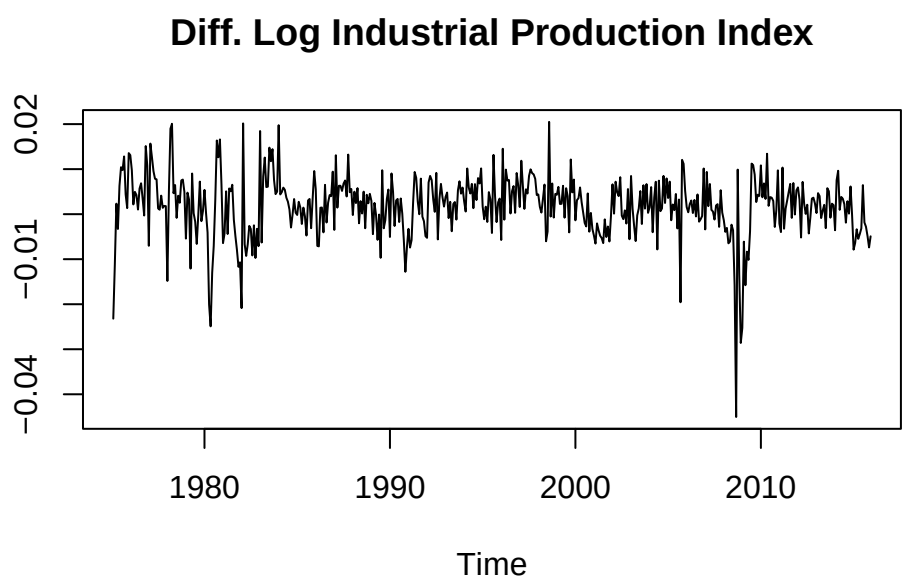

```r
d_usd_ts <- diff(usd_idx_ts)
dl_usd_idx_train_ts <- diff(log(usd_idx_train_ts))


d_ind_ts <- diff(ind_prod_ts)
dl_ind_train_ts <- diff(log(ind_prod_train_ts))

d_opec_ts <- diff(opec_prod_ts)
dl_opec_idx_train_ts <- diff(log(opec_prod_train_ts))
```

Plot

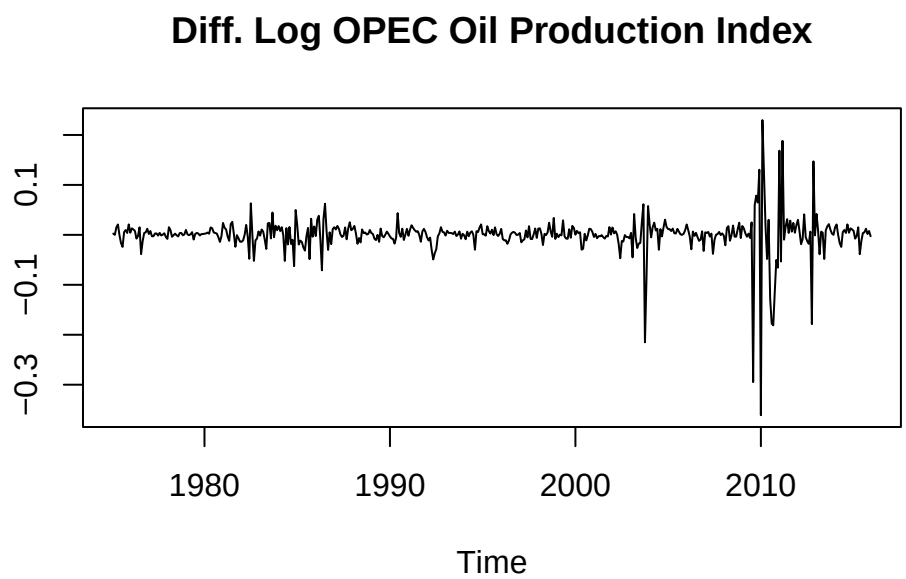
```r
#plot(dl_usd_idx_train_ts, main="Difference U.S. Dollar Index", ylab="")
plot(dl_usd_idx_train_ts, main="Diff. Log U.S. Dollar Index", ylab="")
```




```
#plot(d_ind_train_ts, main="Difference Industrial Production Index", ylab="")
plot(dl_ind_train_ts, main="Diff. Log Industrial Production Index", ylab="")
```

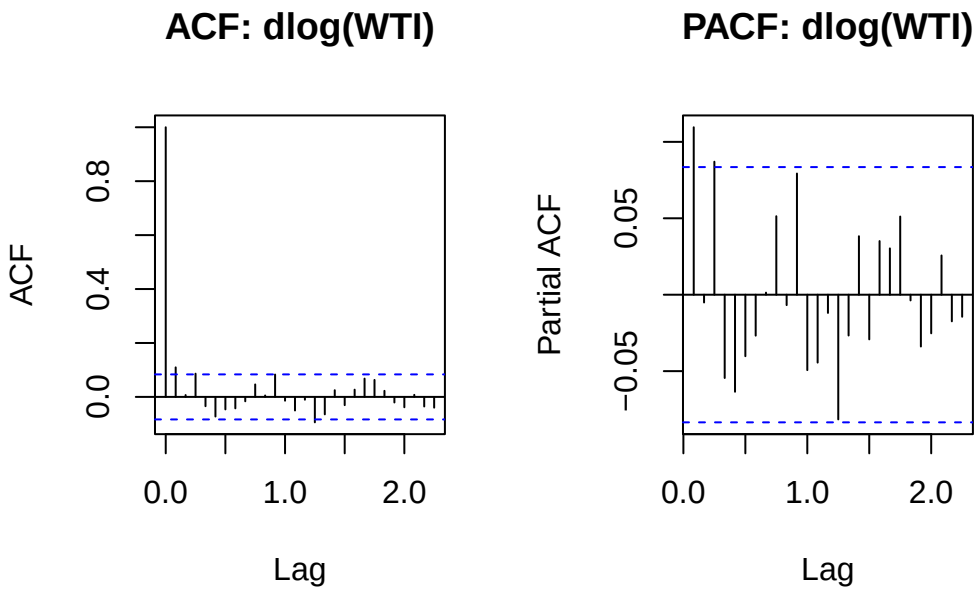


```
#plot(d_opec_idx_train_ts, main="Difference Industrial Production Index", ylab="")
plot(dl_opec_idx_train_ts, main="Diff. Log OPEC Oil Production Index", ylab="")
```



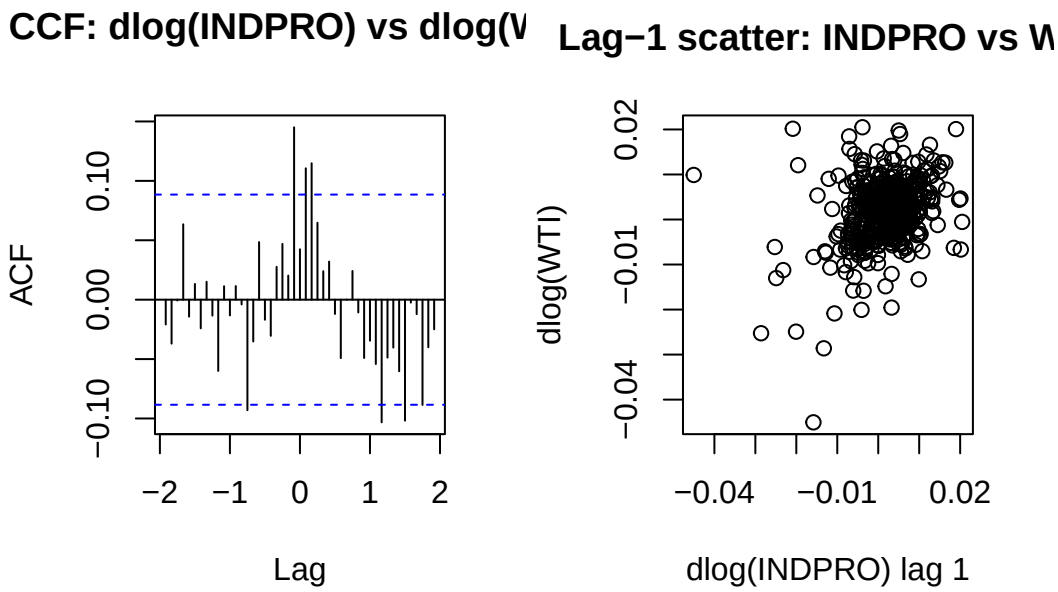
Diagnostics: variogram, ACF, PACF

```
par(mfrow=c(1,2))

acf(dl_crude_oil_train_ts,
    main = "ACF: dlog(WTI)")

pacf(dl_crude_oil_train_ts,
     main = "PACF: dlog(WTI)")
```

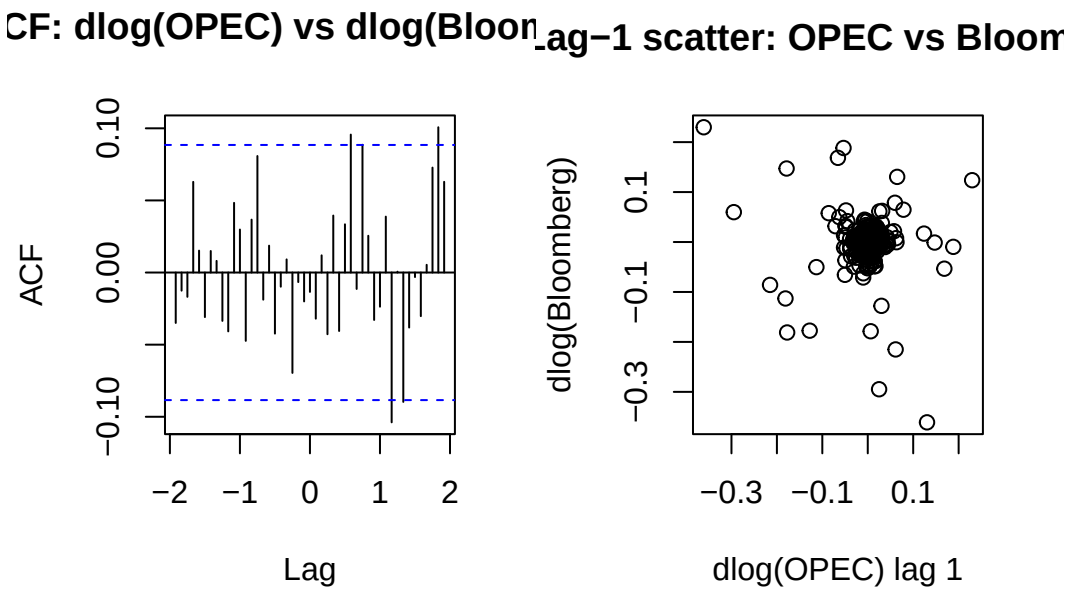


Cross-correlation diagnostics for leading variables

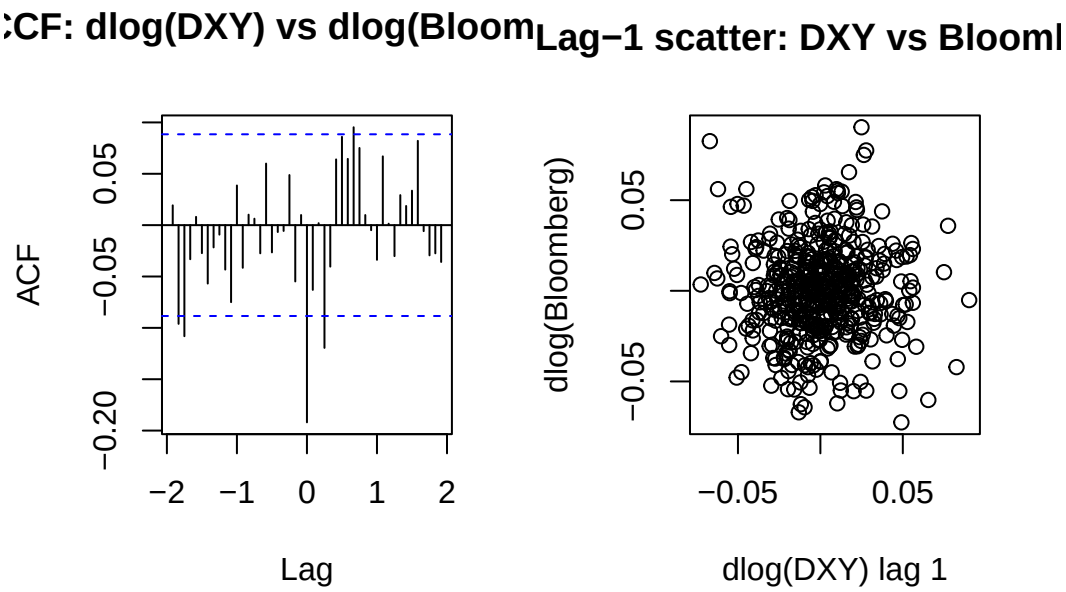
```
par(mfrow=c(1,2))

ccf(dl_ind_train_ts, dl_crude_oil_train_ts, main="CCF: dlog(INDPRO) vs dlog(WTI)")

plot(stats::lag(dl_ind_train_ts, -1), dl_ind_train_ts,
     xlab="dlog(INDPRO) lag 1", ylab="dlog(WTI)",
     main="Lag-1 scatter: INDPRO vs WTI")
```




```
par(mfrow=c(1,2))

ccf(dl_opec_idx_train_ts, dl_crude_oil_train_ts, main="CCF: dlog(OPEC) vs dlog(Bloomberg)")

plot(stats::lag(dl_opec_idx_train_ts, -1), dl_opec_idx_train_ts,
     xlab="dlog(OPEC) lag 1", ylab="dlog(Bloomberg)",
     main="Lag-1 scatter: OPEC vs Bloomberg")
```




```
par(mfrow=c(1,2))

ccf(dl_usd_idx_train_ts, dl_crude_oil_train_ts, main="CCF: dlog(DXY) vs dlog(Bloomberg)")

plot(stats::lag(dl_usd_idx_train_ts, -1), dl_usd_idx_train_ts,
     xlab="dlog(DXY) lag 1", ylab="dlog(Bloomberg)",
     main="Lag-1 scatter: DXY vs Bloomberg")
```



Simple Rules